

X reads keyboard and mouse inputs and makes them available to the applications that need them. Based on the information that the applications get, they perform certain operations and if a result or a change has to be shown on screen, the applications send instructions to X to display it. This is how X services the desktops.

But X by itself does not define how content is displayed on the screen; it only provides a means to display this content. It throws open choices to the client to choose how content is displayed on the screen. When the client asks for a particular action to be done for the display, the server performs it. The X server does not place any restrictions on the client to behave in a particular fashion. In fact, different clients can behave differently using the same X server.

X maintains a list of all its functions in a library file called XLib, using which all data interpretation and visual outputs are handled at the system level. (Basically, “system level” means interacting with the hardware.)

There is a server in charge of visual output and data input, client applications, and a way for them to communicate between each other. The client simply has to tell the server, “give me an area $x*y$ pixels large in the right bottom of the screen”. So, if the client needs data from the keyboard, it would ask X for it rather than monitor the keyboard itself.

4.1.1 Xfree86 and X.Org

The battle in the Open Source community

The X Consortium was formed in 1988 and spearheaded the development and management of the X Server and the tool kits needed for application development. In 1993, the X Consortium, Inc. was formed as a successor to the original MIT X Consortium. In mid-1997, the X Consortium began handing over the responsibility of X to *The Open Group*. The Open Group consisted of vendors who were interested in implementing the X server for Linux applications